FUNCTIONAL REQUIREMENTS DOCUMENT

Project: DeliverEats – Food Delivery App

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Version: 1.1 (revised)

Date: 5 November 2026

This is a simulated Business Analyst portfolio project. No real stakeholders, sign-offs, systems or test executions are involved.

Version History

Version	Date	Revised By	Reason for Change
1.0	28-07-2025	Anandu Priyakumar	Initial FRD aligned with BRD v1.0.
1.1	05-11-2026	Anandu Priyakumar	Revision aligned with BRD v1.1: MVP scope, status models, traceability, UAT status (see change summary).

Revision 1.1 Change Summary

- MVP scope aligned: live GPS tracking, push notifications and the admin dashboard moved to future enhancements (FE-01 to FE-03); real-time tracking replaced by order status milestones.
- Availability target aligned at 99.5% per month in both documents (BRD previously 95%).
- Cash on Delivery decision recorded: not supported in the MVP (RULE-08); checkout wireframe updated.
- Payment status and order status separated into two enumerations with an explicit order lifecycle.
- Rejected orders and failed payments defined: automatic payment reversal (FR-11, UC-14) added; complaints, manual refunds and ratings moved to future enhancements.
- UAT scenarios restated as 'Planned (not executed)' and expanded to 10 scenarios with test cases.
- Traceability repaired: every BR maps to FRs, use cases and UAT scenarios; FR-11 no longer shared by unrelated use cases.
- Fictional-celebrity approvers replaced with role-based simulated approvals, clearly labelled as simulated.
- Terminology standardised: real-time, third-party, As-Is, To-Be, Cash on Delivery.
- Added: in-scope delivery partner wireframe, OTP SMS gateway on the context diagram, business-use-case IDs (BUC-xx).

Document Approvals (Simulated)

All approvals below are simulated for the purpose of this portfolio project. They are role placeholders, not real stakeholder sign-offs.

Approver	Project Role	Approval Type	Date
Anandu Priyakumar	Business Analyst (Author)	Author sign-off	05-11-2026
Mock Project Manager	Project Manager	Simulated approval (portfolio exercise)	05-11-2026
Mock Product Owner	Product Owner (Restaurant side)	Simulated approval (portfolio exercise)	05-11-2026
Mock Technical Lead	Technical Lead	Simulated approval (portfolio exercise)	05-11-2026
Mock Project Sponsor	Project Sponsor	Simulated approval (portfolio exercise)	05-11-2026

Following simulated approval, changes to this FRD would be governed by the project's change management process (impact analysis, review and approval).

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1. Introduction

1.1 Purpose

This FRD defines the functional and non-functional specifications of the DeliverEats MVP, derived from BRD v1.1. It is written for developers, testers, UI/UX designers, solution architects and stakeholders.

1.2 Summary

DeliverEats lets customers browse restaurants, order, pay online and follow order status milestones; restaurants accept or reject and prepare orders; delivery partners collect and deliver them.

1.3 Background

Many local restaurants still take orders by phone and cash, leading to errors and inefficiency. DeliverEats provides one digital workflow for all parties and leaves room for later enhancements such as live GPS tracking, push notifications and an admin dashboard.

2. Project Scope

2.1 In-Scope (MVP)

- Registration with mobile/email and OTP; secure login for customers, restaurants and delivery partners
- Restaurant browsing and digital menus
- Cart and order placement
- Online payment through a payment gateway (Google Pay, Paytm) with automatic reversal on rejection or time-out
- Restaurant order management
- Delivery partner module: accept, pick up, deliver
- Order status milestones for customers
- In-app messages for payment failure and order rejection

2.2 Future Enhancements (not in MVP)

ID	Enhancement	Description
FE-01	Live GPS order tracking	Map-based live location of the delivery partner (replaces status milestones).
FE-02	Push notifications	Push alerts to customers, restaurants and delivery partners (MVP uses in-app messages and OTP SMS only).
FE-03	Central admin dashboard	Admin UI for user management, platform monitoring, complaint handling and reports (MVP uses back-end tools run by the technical team).
FE-04	Complaint handling and manual refunds	Customer complaints, admin review and manually issued refunds.
FE-05	Ratings and reviews	Customers rate restaurants and delivery.
FE-06	Cash on Delivery	Cash payment at the door. Excluded because RULE-02 requires successful payment before the order reaches the restaurant.

2.3 Out-of-Scope

- Loyalty and referral programmes
- Multiple-city delivery support
- Restaurant payroll or HR management
- Voice-based or AI ordering
- In-app chat between customers and delivery partners

- Integration with restaurant inventory systems

3. Functional Objectives / Prioritisation (MoSCoW)

Objective	MoSCoW	Description
Customer registration and secure login	Must	Access control for customers, restaurants and delivery partners
Browse restaurants and menus	Must	Discovery and selection of food
Cart and order placement	Must	Core ordering workflow
Secure online payment, with automatic reversal	Must	Digital transactions; protects customers when orders are rejected
Restaurant accept/reject and preparation status	Must	Restaurant-side control of the workflow
Delivery partner accept, pick up, deliver	Must	Without it no order can be fulfilled
Order status milestones for customers	Should	Transparency without live GPS
In-app payment-failure and rejection messages	Should	Keeps users informed
Live GPS tracking, push notifications, admin dashboard, complaints and refunds, ratings, Cash on Delivery	Won't (this release)	Future enhancements FE-01 to FE-06
Loyalty / referral programme	Won't	Out of scope

4. Context Model and System Overview

4.1 System Context Diagram

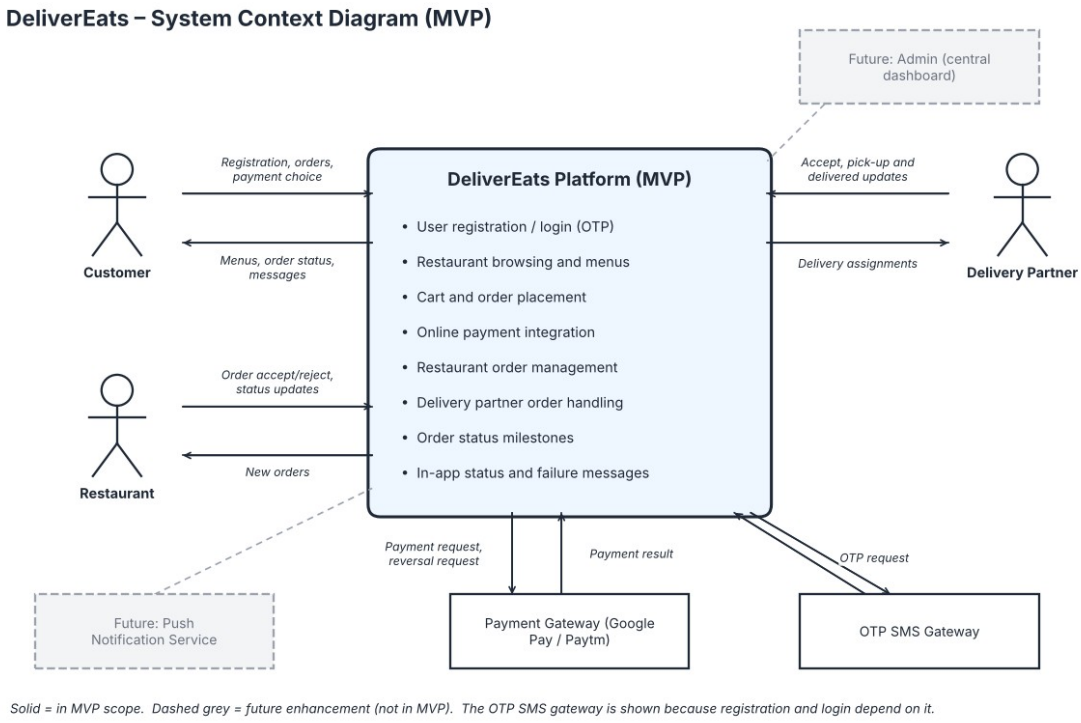


Figure 4.1 – System context (MVP)

4.2 External Interfaces

Interface	Type	Description	Phase
OTP SMS Gateway	Third-party API	Sends one-time passwords during registration (and login recovery).	MVP
Payment Gateway	Third-party payment API	Processes online payments (Google Pay, Paytm) and payment reversals.	MVP

Interface	Type	Description	Phase
Push Notification Service	Third-party API	Push alerts to customers, restaurants and delivery partners.	Future (FE-02)

In the MVP, status and failure messages are shown in-app; only OTP SMS is sent outside the app.

5. Use Case Model

5.1 Use Case Diagram

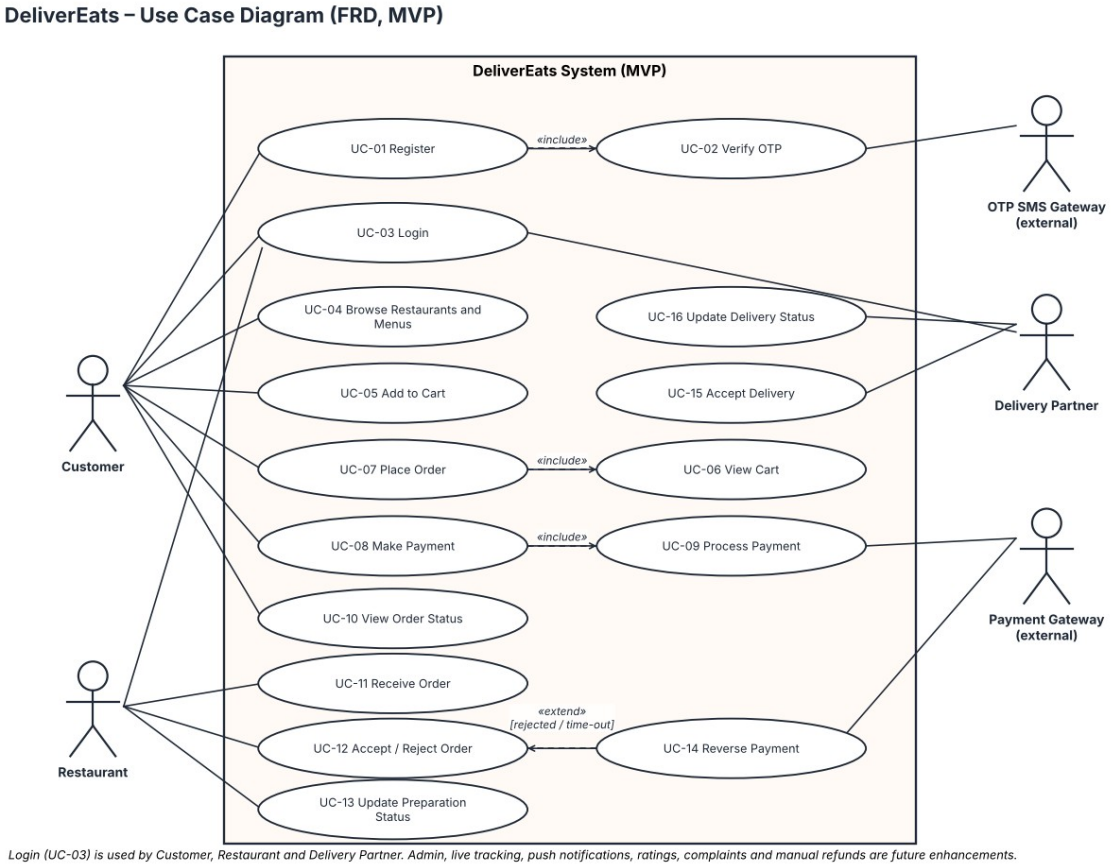


Figure 5.1 – Use cases (MVP)

5.2 Use Case Narratives

UC ID	Use Case	Actors	Preconditions	Postconditions	Flow	FRs	BUC
UC-01	Register	Customer	App installed, internet access	Account created	Enter details > request OTP > verify OTP > account created	FR-01	BUC-01
UC-02	Verify OTP	Customer, OTP SMS Gateway	Registration started	OTP validated	System requests OTP send > user enters OTP > system validates	FR-01	BUC-01
UC-03	Login	Customer, Restaurant, Delivery Partner	Registered account	Access granted	Enter credentials > authenticate > access granted	FR-02	BUC-02
UC-04	Browse Restaurants and Menus	Customer	Logged in	Menu displayed	Browse list > select restaurant > view menu	FR-03	BUC-03
UC-05	Add to Cart	Customer	Logged in, menu displayed	Items stored in cart	Select item > add to cart	FR-04	BUC-04
UC-06	View Cart	Customer	Items in cart	Cart contents displayed	Open cart > view items, quantities, total > edit	FR-04	BUC-04
UC-07	Place Order	Customer	Cart not empty	Order created (Pending Payment)	Review cart > enter address > choose payment method > place order	FR-05	BUC-05

UC ID	Use Case	Actors	Preconditions	Postconditions	Flow	FRs	BUC
UC-08	Make Payment	Customer	Order created	Payment successful or failed	Pay via Google Pay/Paytm > await result > retry or cancel on failure	FR-06, FR-10	BUC-06
UC-09	Process Payment	Payment Gateway	Payment requested	Payment result returned	Receive request > process > return result	FR-06	BUC-06
UC-10	View Order Status	Customer	Order placed	Current milestone displayed	Open order > view status milestone	FR-09, FR-10	BUC-07
UC-11	Receive Order	Restaurant	Payment successful	Order visible on dashboard	Order appears with items, customer, amount	FR-07	BUC-08
UC-12	Accept / Reject Order	Restaurant	Order received	Order Accepted or Rejected	Open order > accept or reject (auto-reject after 5 min)	FR-07, FR-11	BUC-08
UC-13	Update Preparation Status	Restaurant	Order accepted	Status Preparing / Ready for pickup	Select order > update status	FR-07	BUC-08
UC-14	Reverse Payment	Payment Gateway	Order rejected or timed out	Payment status Refunded	System sends reversal request > gateway reverses > status updated	FR-11	BUC-06, BUC-08
UC-15	Accept Delivery	Delivery Partner	Order assigned	Order picked up	View assignment > accept > confirm pickup	FR-08	BUC-09
UC-16	Update Delivery Status	Delivery Partner	Order picked up	Status Delivered	After delivery > tap Delivered > system updates status	FR-08	BUC-09

6. Functional Requirements

Each functional requirement traces to a business requirement in BRD v1.1 and to the use cases that realise it.

FR ID	Description	Priority	Supports BR	Use Cases
FR-01	The system shall allow customers to register with mobile number and email, verified by an OTP sent through the OTP SMS gateway.	High	BR-01	UC-01, UC-02
FR-02	The system shall allow secure login for customers, restaurants and delivery partners.	High	BR-01	UC-03
FR-03	The system shall allow customers to browse restaurants and view their menus.	High	BR-02	UC-04
FR-04	The system shall allow customers to add menu items to a cart and view or edit the cart.	High	BR-03	UC-05, UC-06
FR-05	The system shall allow customers to submit an order from the cart with a delivery address and a payment method.	High	BR-03	UC-07
FR-06	The system shall process payment through the payment gateway (Google Pay, Paytm). The order shall be sent to the restaurant only after payment succeeds. On failure the customer shall be able to retry or cancel.	High	BR-04	UC-08, UC-09
FR-07	The system shall allow restaurants to view incoming orders, accept or reject them within 5 minutes (auto-reject on time-out) and update preparation status (Preparing, Ready for pickup).	High	BR-05	UC-11, UC-12, UC-13
FR-08	The system shall assign accepted orders to an available delivery partner and allow the partner to accept, mark Picked up and mark Delivered.	Medium	BR-06	UC-15, UC-16
FR-09	The system shall allow customers to view the current order status milestone (Placed, Accepted, Preparing, Ready for pickup, Picked up, Delivered).	Medium	BR-07	UC-10
FR-10	The system shall show in-app messages to the user on payment failure and on order rejection.	Medium	BR-08, BR-07	UC-08, UC-10
FR-11	The system shall automatically request a full payment reversal through the payment gateway when an order is rejected or times out.	High	BR-04, BR-05	UC-12, UC-14

6.1 Business Rules Applied

Rule ID	Description
RULE-01	Users must register with a valid mobile number and email, verified by OTP.
RULE-02	Payment must be successfully processed before the order is sent to the restaurant.
RULE-03	The system must immediately show an in-app message on payment failure or order rejection.
RULE-04	Customers must be able to view the status milestone of their active orders in the app.
RULE-05	A delivery partner must handle no more than one active delivery at a time.
RULE-06	A restaurant must accept or reject an order within 5 minutes; otherwise the order is automatically rejected.
RULE-07	When an order is rejected or times out, the full payment is automatically reversed to the original payment method.
RULE-08	Only online payment methods (Google Pay, Paytm) are supported in the MVP. Cash on Delivery is not supported.

7. Data Requirements

MVP-level data elements, with validation rules. Payment status and order status are separate fields.

Field	Description	Type	Validation	Mandatory
Full Name	Customer's full name	Text	Letters and spaces only, max 50 characters	Y
Email Address	Unique email address	Email	Valid email format, no duplicates	Y
Mobile Number	Contact number used for OTP	Numeric (10 digits)	Exactly 10 digits, numerals only	Y
Password	Login password (stored as salted hash)	Hashed text	Min 8 characters with at least 1 uppercase letter, 1 number and 1 symbol	Y
Delivery Address	Delivery location entered by customer	Text	Max 200 characters	Y
Order ID	Unique identifier for each order	Alphanumeric (UUID)	System-generated	Y
Order Total	Total amount payable (INR)	Decimal	Greater than 0, 2 decimal places	Y
Payment Method	Method selected for payment	Enum	Google Pay, Paytm	Y
Payment Status	Status of the payment transaction	Enum	Pending, Successful, Failed, Refunded	Y
Order Status	Current lifecycle status of the order	Enum	Pending Payment, Placed, Accepted, Preparing, Ready for Pickup, Picked Up, Delivered, Rejected, Cancelled	Y
Restaurant ID	Restaurant fulfilling the order	Alphanumeric	System-generated	Y
Menu Item ID	Unique identifier for each menu item	Alphanumeric	System-generated	Y
User Role	Role of the logged-in user	Enum	Customer, Restaurant, Delivery Partner	Y

7.1 Order Lifecycle and Status Model

Payment Status values: Pending, Successful, Failed, Refunded. Order Status values and their usual payment status are:

Order Status	Payment Status	Meaning
Pending Payment	Pending	Order created; awaiting payment result. Restaurant cannot see it yet.
Placed	Successful	Payment succeeded; order sent to the restaurant.
Accepted	Successful	Restaurant accepted within 5 minutes.
Preparing	Successful	Restaurant is preparing the food.
Ready for Pickup	Successful	Food is ready; delivery partner assigned.
Picked Up	Successful	Delivery partner has collected the order.
Delivered	Successful	Delivery partner marked the order delivered.
Rejected	Refunded	Restaurant rejected the order or 5 minutes elapsed; payment reversed automatically (FR-11).
Cancelled	Failed	Customer cancelled after a failed payment; no money taken.

Key transitions: Pending Payment > Placed (payment succeeds) | Pending Payment > Cancelled (payment fails and customer cancels) | Placed > Accepted | Placed > Rejected (rejected or 5-minute time-out, payment Refunded) | Accepted > Preparing > Ready for Pickup > Picked Up > Delivered.

8. Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	The system must respond to user actions within 2 seconds under normal load.
NFR-02	Availability	The platform must achieve at least 99.5% availability per calendar month, excluding planned downtime (about 3.6 hours of unplanned downtime per 30-day month).
NFR-03	Usability	The user interface must support intuitive navigation with a minimal learning curve for all user roles.
NFR-04	Security	Sensitive data must be encrypted in transit (TLS) and at rest. Passwords must be stored as salted hashes, never in plain text.
NFR-05	Maintainability	The platform must be modular so updates and fixes can be deployed with minimal disruption.
NFR-06	Scalability	The system must be able to scale to a 5x increase in users with minimal performance degradation after the MVP.
NFR-07	Compatibility	The app must support Android 10 and above. iOS (13+) may be considered in future releases.
NFR-08	Backup & Recovery	Transactional data must be backed up every 24 hours and recoverable within 2 hours of a failure.
NFR-09	Payment security	No card or UPI credentials are stored by DeliverEats; all payment data is handled by the third-party payment gateway.

9. Assumptions and Constraints

9.1 Assumptions

1. Users have smartphones and stable internet.
2. Restaurant partners will adopt the digital order management system.
3. Delivery partners are available in the initial service area during operating hours.
4. Third-party payment and SMS services work as intended without major integration issues.
5. No legal or regulatory restriction blocks digital orders or payments in the launch region.
6. Stakeholders give timely feedback during requirement finalisation and UAT.
7. Basic administration is done by the technical team with back-end tools until FE-03 is built.

9.2 Constraints

8. MVP supports one city or region.
9. Only Android 10+ is supported in the first release.
10. English only for the MVP.
11. Budget and time exclude: loyalty, in-app chat, voice/AI ordering, multi-city support, inventory integration, live tracking, push notifications and the admin dashboard.
12. No custom payment module; third-party gateways only.
13. No marketing automation, CRM or accounting integration in the MVP.

10. Requirements Traceability Matrix (RTM)

BR > FR > Use Case > UAT scenario. The standalone RTM spreadsheet is in 05_requirements_analysis.

BR ID	Business Requirement	FR ID(s)	Use Cases	UAT
BR-01	Customers must be able to register and log in to the DeliverEats app. Restaurants and delivery partners must be able to log in.	FR-01, FR-02	UC-01, UC-02, UC-03	UAT-01, UAT-02

BR ID	Business Requirement	FR ID(s)	Use Cases	UAT
BR-02	Customers must be able to browse restaurants and view digital menus.	FR-03	UC-04	UAT-03
BR-03	Customers must be able to build a cart and place food orders through the app.	FR-04, FR-05	UC-05, UC-06, UC-07	UAT-04, UAT-05
BR-04	Customers must be able to pay securely online (Google Pay, Paytm). Payment must be returned automatically if an order is rejected or times out.	FR-06, FR-11	UC-08, UC-09, UC-12, UC-14	UAT-05, UAT-06, UAT-08
BR-05	Restaurants must be able to receive, accept or reject incoming orders and update preparation progress.	FR-07, FR-11	UC-11, UC-12, UC-13, UC-14	UAT-07, UAT-08
BR-06	Delivery partners must receive order assignments and update the delivery status.	FR-08	UC-15, UC-16	UAT-09
BR-07	Customers must be able to view the status of their orders in the app (status milestones).	FR-09, FR-10	UC-08, UC-10	UAT-10
BR-08	The platform must inform users in-app of payment failures and order rejections.	FR-10	UC-08, UC-10	UAT-06, UAT-08, UAT-10

11. User Acceptance Testing (UAT) – Planned

These are planned test scenarios. No tests have been executed because no working application exists; detailed test cases and a results tracker are in 06_testing.

UAT ID	Test Objective	Related FR	Acceptance Criteria	Status
UAT-01	Register with mobile/email and verify OTP	FR-01	Account is created; user is able to log in. Invalid OTP shows an error and does not create the account.	Planned (not executed)
UAT-02	Login for each role	FR-02	Valid credentials open the role's home screen; invalid credentials show an error and deny access.	Planned (not executed)
UAT-03	Browse restaurants and view menu	FR-03	Restaurant list is displayed; selecting a restaurant shows its menu categories and items with prices.	Planned (not executed)
UAT-04	Add, edit and view cart	FR-04	Cart shows correct items, quantities and total after each change.	Planned (not executed)
UAT-05	Place order with successful payment	FR-05, FR-06	Payment status Successful; order status Placed; order appears on the restaurant dashboard; Cash on Delivery is not offered.	Planned (not executed)
UAT-06	Failed payment: retry or cancel	FR-06, FR-10	In-app 'Payment failed' message is shown; order is not sent to the restaurant; retry re-attempts payment; cancel sets order to Cancelled.	Planned (not executed)
UAT-07	Restaurant accepts order and updates preparation	FR-07	Order status moves Accepted > Preparing > Ready for Pickup and each change is visible to the customer.	Planned (not executed)
UAT-08	Order rejected or timed out with automatic payment reversal	FR-07, FR-10, FR-11	Both orders become Rejected; payment status becomes Refunded; customer sees an in-app rejection message.	Planned (not executed)
UAT-09	Delivery partner accepts, picks up and delivers	FR-08	Statuses update to Picked Up and Delivered; a partner cannot hold more than one active delivery.	Planned (not executed)
UAT-10	Customer views order status milestones	FR-09, FR-10	Milestones Placed to Delivered display correctly with the current step highlighted; no live map is shown.	Planned (not executed)

12. Glossary

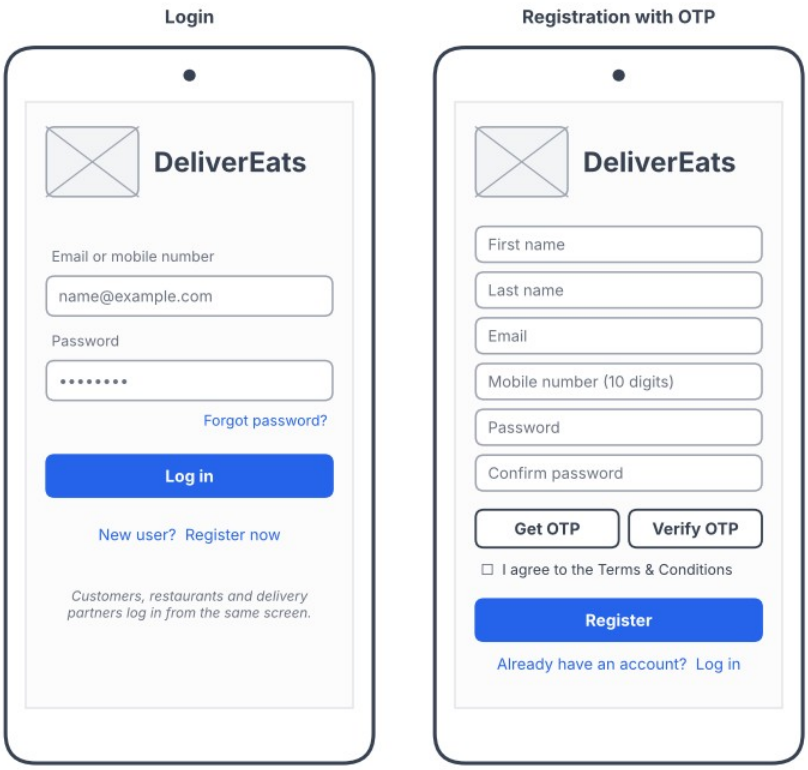
Term	Definition
FRD	Functional Requirements Document.
BRD	Business Requirements Document.
MVP	Minimum Viable Product, the first functional release.
OTP	One-Time Password used to verify identity at registration.
UAT	User Acceptance Testing: validation that the system meets business requirements.
Use Case	A scenario describing how an actor uses the system to reach a goal.
Cart	A temporary container for menu items chosen before ordering.
Delivery Partner	A person who collects food orders and delivers them to customers.

Term	Definition
Order status milestone	A named step in the order lifecycle shown to the customer.
Payment reversal	Automatic return of payment when an order is rejected or times out.
Payment Gateway	A third-party service (e.g. Google Pay, Paytm) that processes online payments.
Restaurant Dashboard	The restaurant's interface for viewing and managing orders and menus.
System Context Diagram	A high-level view of the system and its external actors and systems.
Future enhancement	A feature excluded from the MVP and held in the future-enhancements backlog.

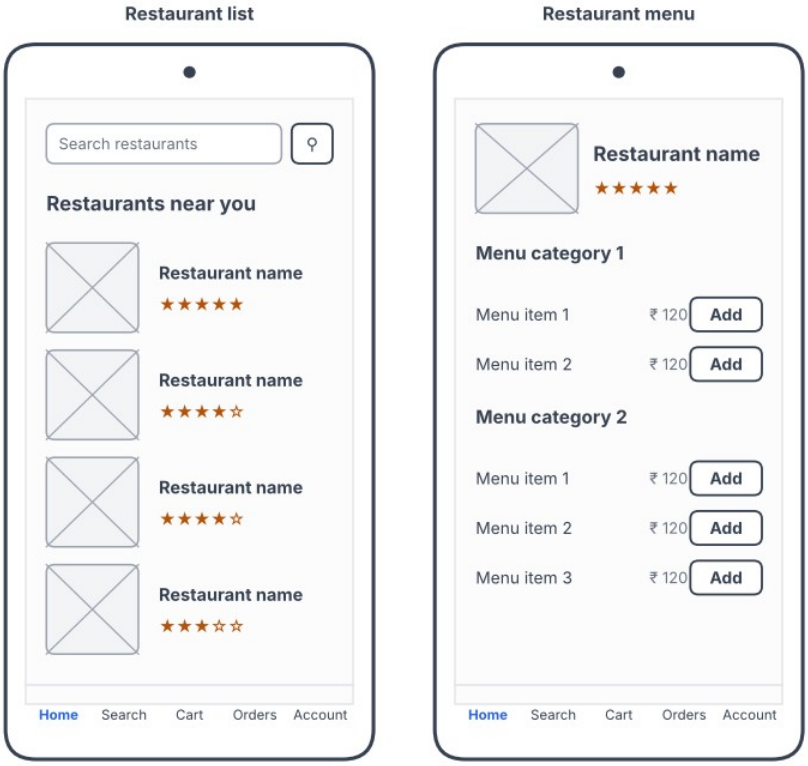
13. Appendices

Appendix	Title	Description
A	Wireframes	Seven MVP screens: login/registration, restaurant list and menu, cart and checkout, payment, order status, restaurant dashboard, delivery partner.
B	UML System Activity Diagram	End-to-end order lifecycle across system components.
C	BRD v1.1 Reference	Source business requirements document.

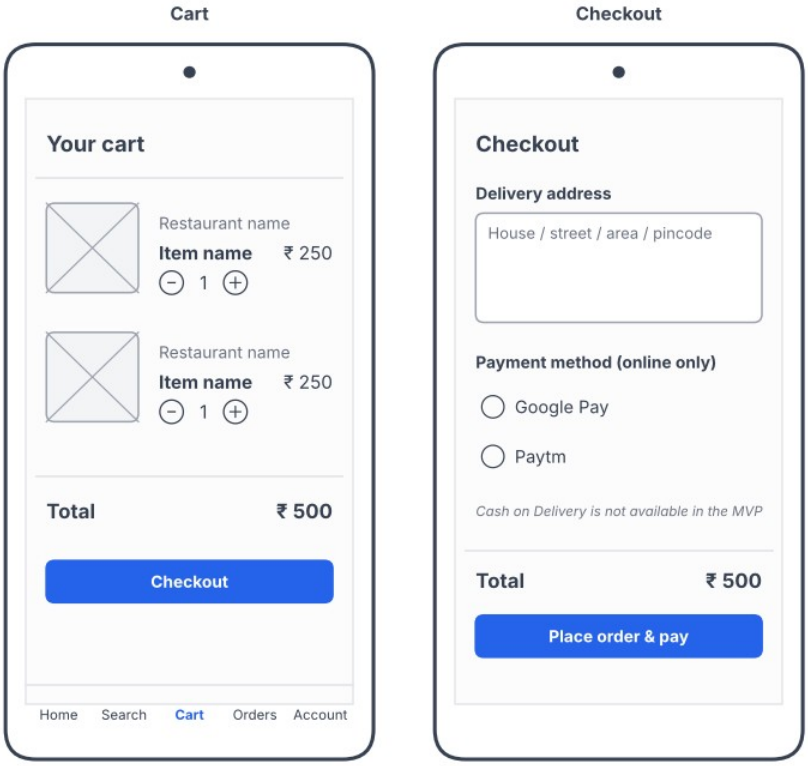
13.1 Appendix A – Wireframes



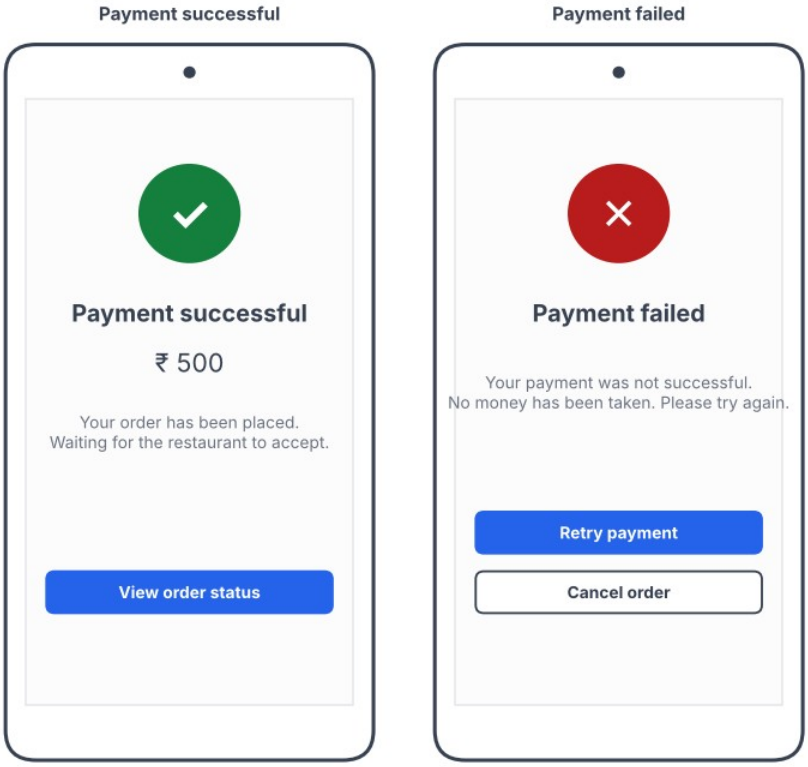
A.1 Login and registration

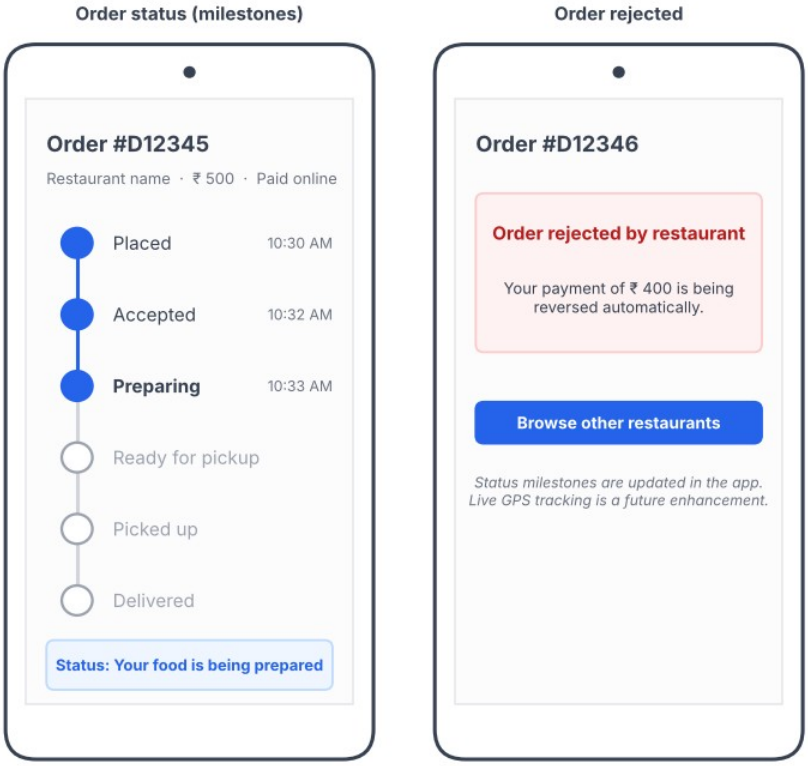


A.2 Restaurant list and menu

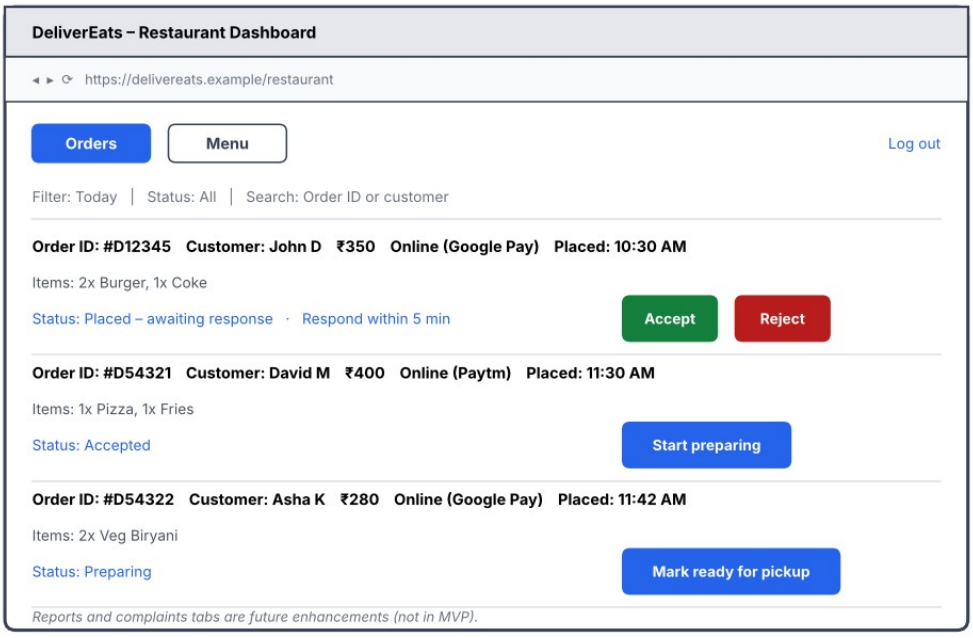


A.3 Cart and checkout (online payment only)

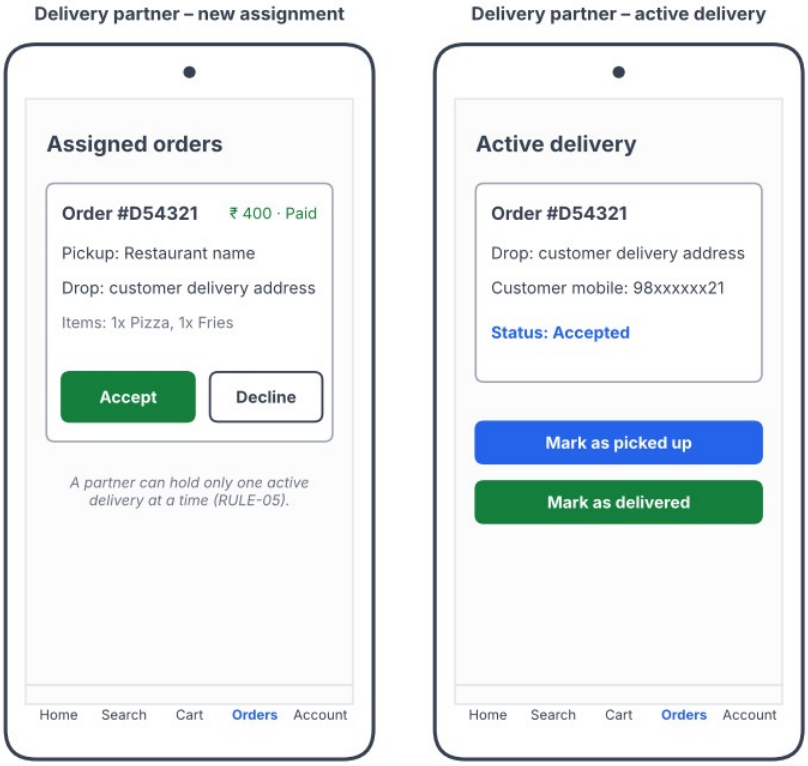




A.5 Order status milestones (live GPS map is a future enhancement)



A.6 Restaurant dashboard (accept/reject and preparation status)



A.7 Delivery partner screens

13.2 Appendix B – UML System Activity Diagram

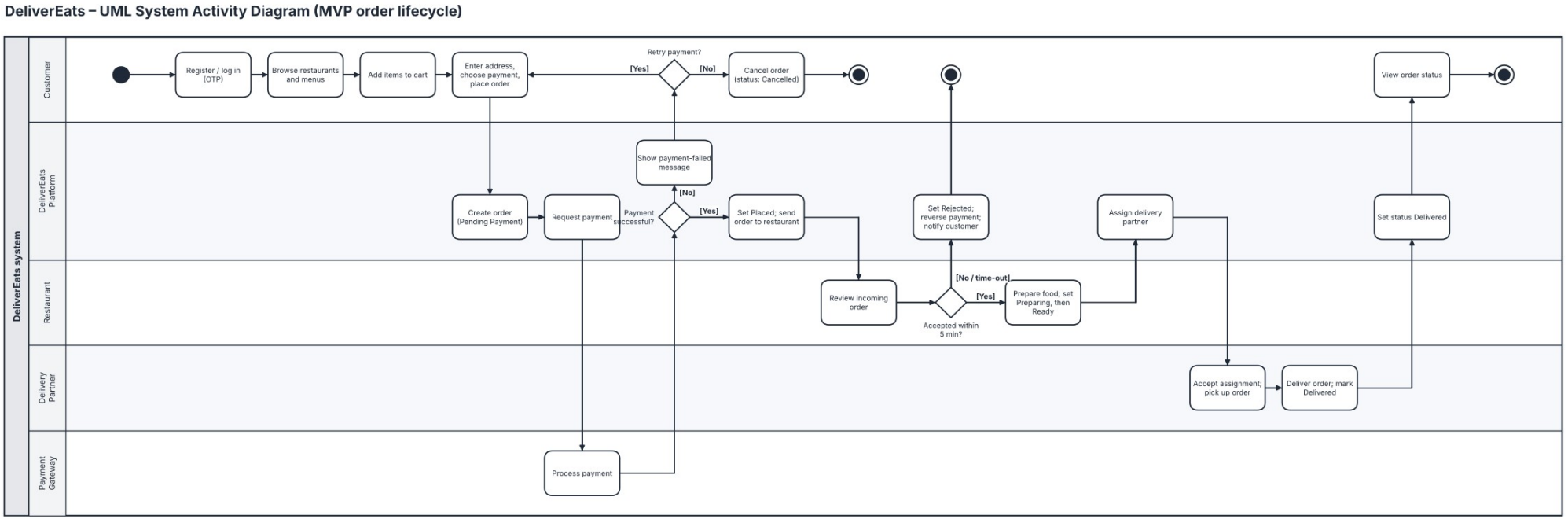


Figure B1 – MVP order lifecycle

13.3 Appendix C – BRD v1.1 Reference

See DeliverEats_BRD_v1.1.pdf in 01_requirements/BRD, which contains the business requirements referenced throughout this FRD.